Engineering Mathematics III

ENGE702

Lecture 15: Laplace Transforms

Laplace Transforms

A Laplace transform takes a function and transforms it into a different looking function.

The purpose of a Laplace transform (for our purposes at least) is that it can be used to transform a differential equation into a form that we can solve algebraically.

This will give us the ability to solve a range of differential equations that we have not previouly been able to solve.

In particular, it is useful when the 'forcing functions', $r(t)$ have steps or sudden changes, for example a switch being applied in an electrical circuit.

Laplace Transforms

A Laplace transform is defined as follows. Given a function $f(t)$, we define the Laplace transform as:

$$F(s) = \mathcal{L}(f(t)) = \int_0^\infty e^{-st} f(t)\ dt.$$

In practice, we will not need to use the definition very much, if at all, since common Laplace transforms can be found in a table.

The original function $f(t)$ is said to be in the **time domain**, and the function is usually denoted with a lower-case letter.

The transformed function, $F(s)$ is said to be in the **Laplace domain**, and the function is usually denoted with an upper-case letter.

Laplace Transforms

We will show a simple example of how to find a Laplace transform, then give a table of other transforms.

Example: Let $f(t) = 1$. Find $\mathcal{L}(f(t))$.

Laplace Transforms

The following table gives Laplace transforms of some common functions:

$f(t)$	$F(s)$
1	$\frac{1}{s}$
t	$\frac{1}{s^2}$
t^n	$\frac{n!}{s^{n+1}}$
e^{at}	$\frac{1}{s-a}$
$\cos(\omega t)$	$\frac{s}{s^2+\omega^2}$
$\sin(\omega t)$	$\frac{\omega}{s^2+\omega^2}$

There are many other 'standard' transforms. There is a table in Section 6.9 of the textbook

Laplace Transforms

There are a number of other rules that will help us to find Laplace transforms of some more complicated functions.

The first is the **sum rule**. If a function consists of two parts that are being added, we can take the transforms separately.

$$\mathcal{L}(f(t) + g(t)) = \mathcal{L}(f(t)) + \mathcal{L}(g(t))$$

The second is the **constant multiple rule**. We can take a constant multiple of a function outside the transform.

$$\mathcal{L}(af(t)) = a\mathcal{L}(f(t))$$

Note that it is **not** the case in general that if a function is the product of two functions we can do the transforms separately.

Laplace Transforms

A final theorem that often comes in useful in practice is known as s-shifting:

$$\mathcal{L}(e^{at}f(t)) = F(s-a).$$

This means that if our function has an exponential term in front, we can take care of it simply by changing every s to $s-a$.

Laplace Transforms

Examples: Find Laplace transforms of the following functions:

1. $f(t) = 5t^2 + 2$
2. $g(t) = e^{3t}\sin(2t)$

Laplace Transforms

In order for the Laplace transform to be useful, we generally need to 'undo' it to get back to the time domain. We do this by taking an **inverse Laplace transform**, so

$$\mathcal{L}^{-1}(F(s)) = f(t).$$

In practice, it is usually a bit trickier finding an inverse transform. We can use the tables when we have to go backwards, but sometimes the function we have is not in the form where we can use the tables directly.

One method that often comes in handy is **partial fractions**. We will now go over a few examples of inverse Laplace transforms.

Laplace Transforms

Example: Find inverse Laplace transforms for the following functions:

1. $F(s) = \dfrac{5s+1}{s^2+25}$
2. $G(s) = \dfrac{6}{(s+1)^3}$
3. $H(s) = \dfrac{1}{s^2+s}$

Exercises

- Find the Laplace transform of the function: $f(t) = te^{3t}$
- Find inverse Laplace transforms of the function: $F(s) = \dfrac{1}{(s-2)^2 + 9}$

Text Book References

Chapter 6.1